

Topic:-

Demonstrate the activities shown on page no. 106 of std IX textbook in front of the students of other classes. Make a video clip and send it to others.

Introduction:-

Accidents, sudden medical emergencies, and natural disasters can occur at any unexpected moment in school environments, playgrounds, or residential neighborhoods. In such critical crisis situations, providing prompt and correct first aid before professional medical help arrives often makes the life-saving difference between recovery and permanent disability.

A crucial component of emergency disaster management is the safe transportation of injured or unconscious patients without worsening their injuries. Incorrect lifting or rough handling can severely damage broken bones or cause irreversible spinal cord trauma. Therefore, every student must learn standardized, scientifically approved emergency rescue techniques.

In our Standard IX science curriculum (Page No. 106), essential practical techniques for transporting injured persons are clearly illustrated. Under the guidance of our science teachers, our team organized a live peer-to-peer demonstration of these vital activities in front of Standard VII and Standard VIII classes. Furthermore, we recorded a clear instructional video clip and shared it digitally with classmates, teachers, and parents.

❖ Objectives of This Demonstration Activity:-

- Mastering the practical rescue and transport techniques illustrated on page 106 of Std IX.
- Demonstrating each method live before junior students with verbal explanations.
- Producing and disseminating an educational video clip for wider community safety awareness.

Aim & Demonstration Setup:-

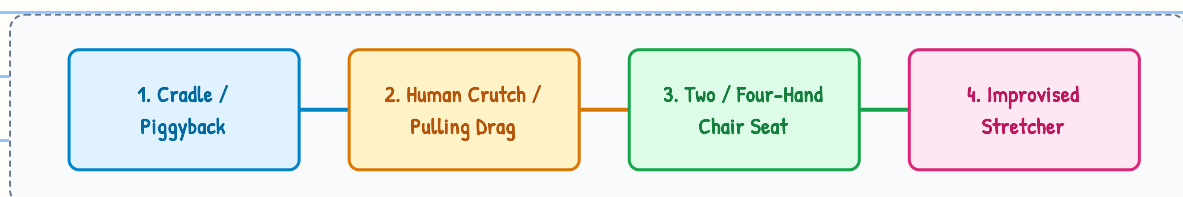
Aim: To demonstrate emergency patient transportation techniques (Std IX page 106) before junior classes, record a step-by-step instructional video clip, and distribute it digitally for community disaster preparedness.

❖ Objectives of the Activity:-

- To practice seven distinct patient rescue methods with precision and rescuer ergonomics.
- To explain indications, contraindications, and safety rules for each transportation technique.
- To film a high-definition video demonstration with clear narration and safety instructions.
- To share the educational video clip across school communication groups and evaluate feedback.

❖ Equipment & Materials Prepared:-

- School gymnasium mats for safe practical demonstrations and simulated rescue exercises.
- Sturdy bamboo poles (2 meters long, 2 units) and thick cotton bedsheets for improvised stretcher.
- First aid demonstration box containing triangular bandages, roller bandages, splints, and sterile gauze.
- Digital smartphone with tripod mount, lapel microphone, and video editing application.



Principles of First Aid & Ergonomics:-

Moving an injured person requires strict adherence to scientific safety protocols.

Rescuers must understand fundamental human anatomy and biomechanical principles to prevent injury to both the patient and themselves.

❖ The Golden ABC Rule of First Aid Assessment:-

- **A - Airway:** Ensure that the victim's airway is clear of vomit, tongue roll, or foreign objects. Keep the neck gently aligned without extreme hyperextension.
- **B - Breathing:** Look for chest elevation, listen for breath sounds, and feel for exhaled air. If breathing has stopped, administer artificial respiration immediately.
- **C - Circulation:** Check carotid or radial pulse. Apply direct pressure with sterile pad to control any severe external bleeding before attempting transport.

❖ Rescuer Biomechanics & Safe Lifting Ergonomics:-

- Keep your spine straight and bend your knees; lift loads using thigh muscles, never back muscles.
- Hold the patient's weight as close to your own body's center of gravity as possible.
- Communicate clearly with partner rescuers using synchronized count signals: "1, 2, 3, Lift!".
- In suspected spinal or neck trauma cases, never bend the patient's neck or body sideways.

Adhering to these principles ensures that emergency transport remains fast, safe, and controlled.

Patient Transport Methods (Page 106):-

Page No. 106 of the Standard IX textbook illustrates seven practical rescue techniques. Below is our systematic operational breakdown demonstrated during the session:

❖ Single Rescuer Transportation Techniques:-

- **1. Cradle Method:** Suitable for light-weight children or small individuals. The rescuer slides one arm behind the child's back and the other beneath the knees, lifting smoothly.
- **2. Carrying on Back (Piggyback):** Ideal for conscious victims who can support their torso and hold onto the rescuer's shoulders. The rescuer supports the victim's thighs.
- **3. Human Crutch:** For conscious victims with one injured leg or ankle. Rescuer stands on the injured side, places victim's arm over rescuer's shoulder, and holds victim's wrist and waist.
- **4. Pulling / Dragging Method:** Used in smoky rooms or low tunnels. Rescuer pulls the victim along the floor by holding cloth at shoulders, keeping head safely supported near ground level.

❖ Two-Rescuer Transportation Techniques:-

- **5. Two-Hand Chair:** Rescuers grasp each other's wrists to form a seat while other arms support patient's back; used for patients who cannot stand but can hold rescuers.
- **6. Four-Hand Chair:** Rescuers grasp their own and each other's wrists in a locked four-hand square to carry a heavier conscious patient safely.
- **7. Improvised Stretcher:** Two bamboo poles threaded through a strong blanket or buttoned shirts to create a rigid, stable cot for unconscious patients.

Live Demonstration Session Details:-

The live demonstration was conducted on 18th August in the School Activity Hall before eighty students of Standard VII and Standard VIII, supervised by our science teachers and physical training instructor:

<i>Method Name</i>	<i>Patient Condition</i>	<i>Rescuers</i>	<i>Key Precaution Emphasized</i>
Cradle Method	Child / Light weight	Single Rescuer	Support neck & back; avoid if fracture exists
Piggyback Carry	Conscious patient	Single Rescuer	Patient must hold rescuer's shoulders firmly
Human Crutch	Single leg sprain	Single Rescuer	Rescuer always stands on patient's injured side
Dragging Method	Fire / Smoke zone	Single Rescuer	Keep body low beneath toxic smoke layer
Four-Hand Chair	Heavier conscious	Two Rescuers	Lock wrists securely; step forward synchronously
Blanket Stretcher	Unconscious patient	Two to Four	Check fabric strength before lifting off ground

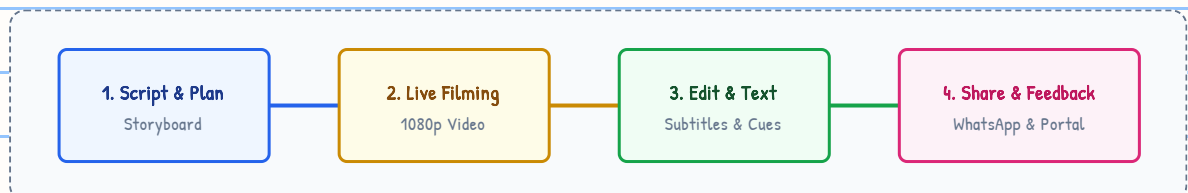
Junior students actively participated in mock trials under close guidance after our demonstration.

Video Clip Production & Sharing:-

To ensure the emergency rescue knowledge reached a wider student community and families, we filmed, edited, and distributed a comprehensive instructional video clip:

❖ *Production Stages of the Educational Video:-*

- **Scripting & Planning:** Prepared a 7-minute structured storyboard explaining the purpose, step-by-step actions, and precautions for each of the seven rescue techniques.
- **Filming Session:** Recorded in high definition (1080p) using tripod stabilization, multi-angle camera shots (close-up of hand locks), and clear voiceover narration.
- **Post-Production Editing:** Added subtitle text, highlighted caution boxes, on-screen timers, and emergency helpline numbers (108 Ambulance, 101 Fire Brigade).



The video clip was shared via official Class WhatsApp groups, school Google Classroom, and teachers' forums.

Technical Evaluation & Trial Analysis:-

During practice trials and live student demonstrations, our team evaluated critical technical parameters including rescue speed, rescuer physical fatigue, and patient safety comfort:

❖ Comparative Technical Observations:-

- **Improvised Stretcher Assembly Time:** With two trained rescuers, an improvised blanket-pole stretcher was fully assembled and tested within 58 seconds.
- **Load Distribution in Chair Methods:** The Four-Hand Chair distributed patient weight evenly across four wrists, allowing rescuers to carry 45 kg over 30 meters with minimal strain compared to Two-Hand Chair.
- **Cervical Neck Stability in Dragging:** Keeping the pull angle low ensured that the simulated patient's cervical spine did not bend, maintaining safe body alignment.

❖ Key Mistakes Identified and Corrected During Practice:-

- **Initial Error:** Lifting patients with arched backs caused lower back muscle discomfort.
Correction: Instructed rescuers to squat, keep backs straight, and lift with quad muscles.
- **Initial Error:** Rescuers stepping with opposite feet caused unstable wobbling.
Correction: Synchronized stepping: both rescuers always stepped forward with inside feet first.

Societal Importance & School Safety:-

Emergency first aid training is not merely an academic syllabus exercise; it is an indispensable life skill that fosters responsible citizenship, empathy, and collective community safety.

❖ Real-Life Applications in Everyday Situations:-

- **School Sports Day & Playgrounds:** Immediate assistance for sprained ankles or fractures during athletic events using the Human Crutch or Two-Hand Chair methods.
- **Domestic Accidents & Fire Incidents:** Rapid evacuation through smoke-filled stairwells using the Clothes Dragging method while remaining beneath toxic fumes.
- **Monsoon Floods & Natural Calamities:** Safely moving elderly neighbors and children across waterlogged areas using improvised stretchers and piggyback carries.

❖ Recommendations for School Safety Preparedness:-

- Conducting mandatory hands-on first aid practical workshops at the start of every academic year.
- Displaying laminated illustrated posters of Page 106 rescue methods in school corridors and gym.
- Forming trained student "Emergency First Responder Squads" across all high school divisions.

Empowering youth with emergency rescue skills creates safer, resilient communities capable of immediate action.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Mastered practical patient rescue and transportation techniques with confidence and correct posture.
- Enhanced communication, teamwork, and presentation skills through live student demonstrations.
- Acquired digital media literacy skills by scripting, shooting, editing, and sharing an instructional video.

❖ Project Conclusion:-

Our project successfully demonstrated all the patient transportation activities shown on Page 106 of the Standard IX textbook in front of junior students. The demonstration generated tremendous enthusiasm and practical understanding among peers. The instructional video clip reached over 200 students and parents, fulfilling our social responsibility of spreading life-saving disaster management awareness.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard IX Science & Technology (Page 106, Environmental Management).
- Indian Red Cross Society – *First Aid and Emergency Care Field Manual*, New Delhi.
- National Disaster Management Authority (NDMA) – *School Safety Policy Guidelines*, Govt. of India.
- St. John Ambulance Association – *First Aid Training Manual for Schools*.